

Call Analysis – Features and Components / Dependencies

Phase key: **Initial** = part of the first release as discussed; **Later** = discussed as a follow-on or aspirational; **Open** = needs a decision or definition before it can be planned.

1. Features

ID	Feature	Description	Components	Phase
F1	Continuous categorization of every interaction	Analyze and persist every incoming interaction as it arrives, piggybacking on Call Summarization; results tied to the individual interaction.	C1, C3, C5	Initial
F2	Seed taxonomy (category / subcategory)	Team drafts a standard dental-insurance taxonomy; reviewed by Ashley, Carmen and Linda before use. Possibly separate sets for member vs. provider, and voice vs. chatbot.	C2	Initial
F3	Category set definitions by payer / plan / group / subgroup	Associate a category set with the level it applies to and its identifier, so standard sets run on all interactions and client-specific sets (e.g. CareSource Ohio) run only where they apply.	C2, C6	Initial
F4	Versioned category sets	Sets are versioned; each finding records the version it was classified under so reports can distinguish periods. Category rows are never edited in place.	C2, C5	Initial
F5	Per-subcategory drill-down extraction	Beyond subcategory, extract what makes the interaction actionable. Examples: Benefits – procedure vs. maximum/deductible, which procedure code; Find a dentist – known vs. new dentist near home, list emailed vs. read by phone; Toolkit – password reset failures, MFA code slow/missing. Start with benefits/eligibility, review specs with the business, iterate.	C4, C5	Initial
F6	Association with interaction metadata	Results queryable by caller type, group, subgroup, payer/plan, source/channel, queue, state, commercial vs. government programs, individual.	C6	Initial
F7	Canned per-group report	Category/subcategory percentages per group, in a form Chris can send to customers.	C9	Initial
F8	CareSource Ohio PCIR / PCDR reports	Generate the monthly provider inquiry and provider dispute reports from customer-supplied category sets using the same framework, replacing the manual monthly process. Funding not approved by GP – confirm whether in scope.	C2, C9, C6	Open
F9	Data externalized and joinable	Results live in our own Oracle database, joinable to enterprise data (group, subgroup, member), not locked in Genesys. Table design reviewed with the team and reporting engineering before build.	C5, C6, C10	Initial
F10	Evolving extraction and reprocessing	Extraction attributes and analysis logic can change over time; interactions can be reprocessed by criteria (date range plus category/set/client filters). Needs a tighter definition of what "evolve" covers.	C4, C8	Open (definition)
F11	Category discovery	AI clustering surfaces new categories/subcategories not in the seed list, on a periodic cadence (undecided).	C7, C11	Later
F12	Category curation by the business	Business users edit discovered categories and define new or separate ones through an admin UI, without a feature-team change.	C11, C2	Later
F13	Standard BI tool support	Power BI / Tableau can consume the analytics data.	C10	Later
F14	Ad-hoc data mining	Users run their own queries: keyword search (e.g. crowns for client ABC), time-boxed questions (interactions mentioning a mailer since it was sent).	C12	Later
F15	Conversational analytics	Chat with the data; the system aggregates on the fly (OpenAI financial-integration analogy). Approach to be researched.	C12	Later

ID	Feature	Description	Components	Phase
F16	Dashboards	Dashboards over the analytics tables once reports exist.	C9, C10	Later
F17	Misroute and cross-plan analysis	Original queue vs. group the interaction was actually for; how many interactions are for other Delta Dental plans (transfer volume).	C6; Genesys queue data	Open (data availability)
F18	Call-flow / routing change effectiveness	Measure impact of IVR and routing changes on volume and deflection. Needs a concrete definition before planning.	C5, C6, C9	Open (definition)
F19	Predictive use for conversions	Learn from past large-group conversions (e.g. TenCare dental home → 50% provider-finding calls) to forecast or offset volume for the next one.	C5, C6, C12	Later
F20	Additional channels	Extend to member chatbot sessions, e-mail, other interaction types (outbound?, mail?). For the chatbot, also record whether the request was fulfilled.	C1, C3, C6	Later / Open (channel list)
F21	Client opt-out exclusion	Skip AI processing for plans/groups that do not approve it (e.g. Louisiana).	C13	Initial
F22	Case vs. interaction anchoring	Decide whether findings attach to the interaction or to the case; multi-case interactions require an extra LLM pass and raise cost.	C14, C5	Open (decision)
F23	Genesys roadmap check	Get a demo/date for Genesys native analytics; only relevant if its data can be exported. Decision so far: proceed regardless.	—	Open (info)

2. Components

ID	Component	Description	Dependencies
C1	Summarization pipeline hook	Extension point in the existing Call Summarization service: after an interaction is summarized, trigger categorization/extraction and persist results. Runs continually as interactions arrive.	Existing Call Summarization service; Genesys transcript feed; TRANSCRIPT / TRANSCRIPT_SUMMARY tables
C2	Category configuration store	Defines what we are looking for: ANALYTIC_CATEGORY (name/value/desc, parent for category→subcategory), ANALYTIC_CATEGORY_SET (named, versioned set), ANALYTIC_CTG_VERSION (set-version ↔ category association), CATEGORY_SET_CONF + CATEGORY_TYPE (applies a set to payer / plan / group / subgroup / member and its identifier). Categories are immutable; changes create new rows and a new set version.	Oracle; data-model review with reporting/warehouse engineering (Wendy); decision on active/inactive flag vs. latest version
C3	Categorization service	LLM classification of each interaction against the active category set(s) that apply to it (standard set for every interaction, plus client-specific sets such as CareSource Ohio when the applicability rules match).	C1, C2, C6, C13; LLM access (Azure OpenAI); prompt templates
C4	Attribute extraction service	Per-subcategory "drill-down" extraction (e.g. procedure code, deductible vs. maximum, list emailed vs. read by phone, MFA code delay). Extraction spec is configurable per subcategory and evolves over time.	C3; extraction-spec storage (no table in the current ER diagram — open); business review of specs per subcategory
C5	Analytics results store	What we found: CATEGORY_ASSIGNMENT (case → category version, raw detection data) and TRANSCRIPT_ANALYTIC_CTG (found categories with plan / client / subclient identifiers). Each finding points to the category version row it was classified under.	C2; C6; Oracle
C6	Interaction metadata linkage	Joins results to interaction attributes: caller type (member/provider/broker), group, subgroup, payer/plan, source/channel, queue, state, commercial vs. government programs, individual. Assumed available on the interaction for group/subgroup; queue depends on Genesys.	INTERACTION, CASE, INTERACTION_CASE tables; Genesys export of queue data (open); case-mapping resolver (C14) where an interaction spans several cases
C7	Category discovery / clustering job	Periodic clustering of historical interactions to surface categories/subcategories not in the seed list; feeds curation (C11) and new set versions (C2).	C5 or transcript store; LLM/embedding service; batch scheduling
C8	Reprocessing job	Re-run categorization/extraction for interactions matching a criteria (date range, category, set, client, etc.) after definitions change. Policy so far: new set versions apply to new interactions; reprocessing is explicit and criteria-driven.	C3, C4, C5; job scheduler; LLM cost budget
C9	Static reporting	Canned, well-structured reports: per-group category/subcategory percentages for customers; CareSource Ohio PCIR and PCDR monthly reports.	C5, C6; CareSource category sets in C2; PCDR disputes flagged by CSR (RCST dropdown); PCIR from the two provider queues
C10	BI integration	Views / warehouse feed so Power BI or Tableau can be pointed at the analytics data; requires key-value typed identifiers in CATEGORY_SET_CONF to be interpretable downstream.	C5, C6; data warehouse / reporting engineering
C11	Category administration UI	Business-facing screens to curate categories and subcategories: edit discovered ones, define new or separate ones, manage sets and versions, without a feature-team change.	C2; authentication; audit
C12	Ad-hoc query and conversational analytics	Data-mining capability: keyword search, user-defined filters and time windows; later a chat interface that aggregates on the fly from the analytics data.	C5, C6, C10; LLM with data access; research on approach
C13	Data governance and exclusions	PII/PHI handling on approved storage; client opt-out (e.g. plans that do not approve AI processing) excluded from analysis.	Security review; list of excluded plans/groups

ID	Component	Description	Dependencies
C14	Case-mapping resolver	Extra LLM step that decides which case a finding belongs to when one interaction spans multiple cases (provider calling about several members; inquiry + dispute in one call). Needed only if results are tied to case rather than interaction.	C3; CASE / INTERACTION_CASE; decision on case vs. interaction anchoring (open)

3. Open items to resolve before planning

ID	Open item	Affects
O1	Where per-subcategory extracted attributes are stored (no table in the current ER diagram; parent_transcr_anal_ctg_id was described as "details of what we found" but not decided).	C4, C5
O2	Findings anchored to interaction vs. case; whether the case-mapping resolver is in scope.	C14
O3	Which specific reports each release is intended to support.	C9
O4	Definitions for "effectiveness of call-flow and routing changes" and "evolving attributes / analysis logic".	F10, F18
O5	Availability of original-queue data from Genesys.	C6, F17
O6	Active/inactive indicator on category sets vs. latest-version lookup.	C2
O7	Channel list for later phases (outbound, mail).	F20
O8	Separate category sets for member vs. provider and voice vs. chatbot.	F2
O9	Group/subgroup on the interaction – assumed available; confirm against the interaction → case → member → enrollment → plan path.	C6